

General biology ඉල්ලිය session 02.

Eukaryotic Cell

* ආසාදන ආකාරයේ ඇත.

They are highly evolved cells processing specialized organelles, a definite membrane bounded nucleus and highly organized cytoplasm.

* සෛලයක් ලෙස ජීවත්වන දෑ සෛලයේ ඇති විවිධ ව්‍යුහයන් වන බැවින් සෛලයක් ලෙස සැලකිය යුතුය.

* සෛලයේ ඇති ව්‍යුහයන්, ඉන්ද්‍රියාංග, වර්ගයන් විවිධ වේ.

protoplasm - ජීවී ජලාස්මය

protoplasm has two parts

1. Cytoplasm - සෛල ජලාස්මය.
2. Nucleus - න්‍යෂ්ටිය

* Cytoplasm has number of compartments
සෛල ජලාස්මය තුළ ඉතිරි වේ.

* They are called as organelles (ව්‍යුහයන්) and carry out different functions.

* These organelles may or may not be bound by a membrane (පටලය).
ex: Mitochondria, Ribosome, Golgi apparatus, Lysosome.

- nucleolus - නූക്കുලිය, nucleus - නූക്കുලිකාව
- ER - ദുർഭ: ദ്രോശരണ ජාලය.
- Rough ER / Smooth ER, lysosome.
(no ribosomes).

සෛල ද්‍රව්‍යමය දැක්වීම කිහිපයක් මේ වන දුර්වාරයන් දුරකණ.

Types of Eukaryotic cell

1. Plant cell - ශාක සෛල
2. Animal cell - සතු සෛල.

Plant cell	Animal cell
* contains cell wall	* does not contain cell wall
* contains a large central vacuole (මහලු ආසාදිත තේරුමක් ඇත).	* contains number of small vacuoles (නුතන නුතන මේරුමක් ඇත).
* contains a chloroplast (මේරුමක් ඇත)	* does not contain chloroplast (මේරුමක් නැත).

සතු සෛලයන් හා ශාක සෛලයන් දැක්වීම බලා ගන්න.
cell wall / chloroplast / Large central vacuole

Structure of the Plasma membrane

ද්‍රව්‍යමය පටලයේ ව්‍යුහය.

* ද්‍රව්‍යමය පටලය සෑදීම දුර්වල ලෙස හා ස්ථිරව පවතී.

ලිපි layer දෙක අතර Hydropobic වලයක් ඇත.
Hydropobic හා ලිපි layers තවම වටහා නැත.

* ഈ പ്ലാസ്മ കട്ടി പരമാവധി 20 മൈക്രോമീറ്റർ വരെ ഉണ്ടാകാം.

* Some protein molecules are small and only a part of project at the surface.

ചെറു പ്രോട്ടീൻ മോളിക്യൂൾ ആയതിനാൽ അവ പ്ലാസ്മ കട്ടിയിൽ പറ്റിയിരിക്കാം.

Extrinsic / peripheral protein

* They are superficially attached to the lipid layer

ആന്തരികമായി attach ചെയ്തതല്ല (not deep).

* They ~~are~~ can be easily removed (ആസാൻ ചെയ്യാൻ എളുപ്പം)

*

* Carbohydrates forms the antennae projecting on the surface.

പ്രോട്ടീൻ മോളിക്യൂൾ പ്ലാസ്മ കട്ടിയിൽ പറ്റിയിരിക്കാം.

* The cells ~~are~~ recognize one another by these.
 ഈ ~~are~~ കോശങ്ങൾ തമ്മിൽ പറ്റിയിരിക്കാം.

cell wall (കോശ ഭിത്തി)

* i) primary cell wall

→ is laid on the middle lamella within the cell
 (അതിൽ ഉൾക്കൊള്ളുന്നു)

→ It consists of cellulose microfibrils running through the matrix. (അല്ലെങ്കിൽ കട്ടി)

No: _____
* The most favoured model is the fluid mosaic model (ଅବସ୍ଥାପନ ମଡେଲ)

* Proposed by Singer and Nicholson in 1972

* Plasma membrane is made up of lipid, proteins and carbohydrate matrix.

* Lipids form a fluid bilayer (ଅବସ୍ଥାପନ ଦ୍ୱିତୀୟ)

* In this bilayer, heads of lipid molecules are directed outwards and they are polar and hydrophilic

ତେଜ ଦିଗରେ ଅଛି, ଡିଜିଟାଲ କରନ୍ତୁ ଏବଂ ଏହା ଧୂଳି
ସହ (ଅବସ୍ଥାପନ) କିମ୍ବା ପ୍ରତିଷ୍ଠା କରନ୍ତୁ.

* Tails are directed inwards (ଅବସ୍ଥାପନ) and they are non polar and hydrophobic.
(ଅବସ୍ଥାପନ) (ଅବସ୍ଥାପନ)

Proteins are two types

(1) Intristic/Intergal (2) Extrinsic/peripheral
(ଅବସ୍ଥାପନ)

Intrinsic / Intergal protein.

* embedded bilayer (ଅବସ୍ଥାପନ ଦ୍ୱିତୀୟ)

* They are held by hydrophobic and hydrophilic parts of lipid molecules and cannot be easily dislodged. (ଅବସ୍ଥାପନ) (ଅବସ୍ଥାପନ).

(ii) Secondary cell wall

It is build up by addition of cellulose layers on the inside surface of primary cell wall

Plasmodesmata (പ്ലാസ്മൊഡെമറ്റ)

• primary cell wall and secondary cell wall

പ്രാഥമിക കോശ ഭിത്തിയും ദ്വൈതീയ കോശ ഭിത്തിയും തമ്മിലുള്ള ചെറിയ തുളക്കൾ പ്ലാസ്മൊഡെമറ്റാകുന്നു

• The cell wall does not form a continuous layer and ~~is~~ is interrupted by minute pores called plasmodesmata.